StyleMorph: Disentangled 3D-Aware Image Synthesis with a 3D Morphable StyleGan

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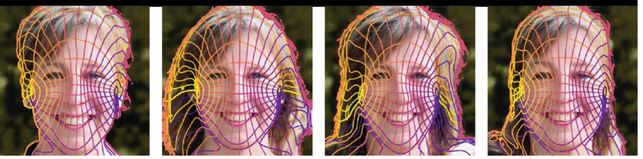
https://stylemorph.github.io/stylemorph/

Motivation

Our objective is to fully **disentangle** 3D **shape**, **viewpoint**, **object** and **background** generation on photorealistic synthesis.

Contribution



Photorealistic modelling with a 3D **morphable** model from 2D image supervision. State-of-the-art **disentangled** control over **shape**, **viewpoint**, **object** and **scene** appearance.

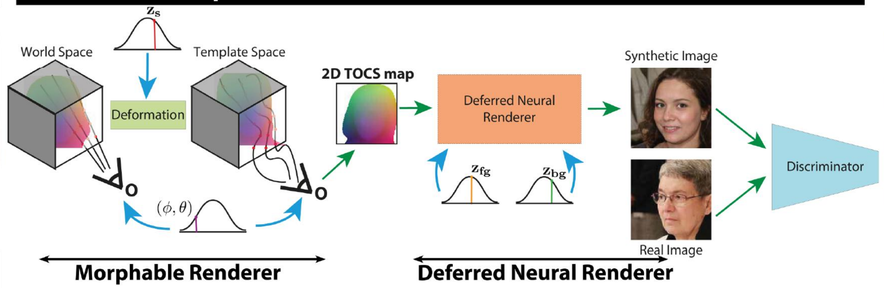
1. Introducing **Template Object Coordinates** (**TOCS**) as a powerful **deformation-equivariant** descriptor of 3D shape;

2. Conditioning a 2D style-based **DNR** with TOCS map for shape and appearance disentanglement.

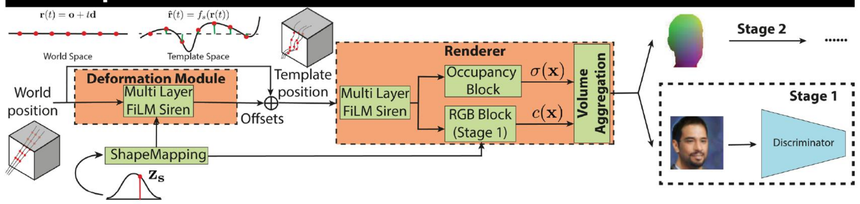
Comparison with SOTA

		Disentanglement		Template	Unposed
		Shape	Scene	-	-
Low-resolution	GRAF	✓	✗	✗	✓
	GIRAFFE	✓	✓	✗	✓
	pi-GAN	✗	✗	✗	✓
	Disentangled3D	✓	✗	✓	✓
Photorealistic	StyleNERF	✗	✗	✗	✓
	CIPS3D	✓	✗	✗	✓
	StyleSDF	✗	✗	✗	✓
	EG3D	✗	✗	✗	✗
	GiraffeHD	✓	✓	✗	✓
	Ours	✓	✓	✓	✓

Overall Pipeline



Morphable Renderer

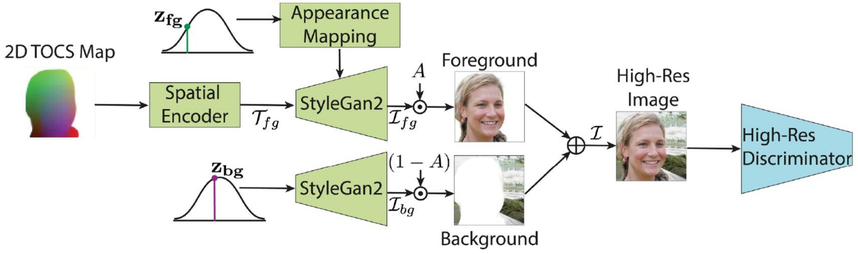


A SIREN network learns to offset **world** points $\mathbf{r}(t)$ to **template** space $\hat{\mathbf{r}}(t)$. This allows to express each instance as a **deformation** of an **instance-agnostic** template shape.

The TOCS Map, the object **coordinates** in **template** space $\hat{\mathbf{r}}(t)$ is rendered to capture all and only the **3D geometrical** aspects of image formation to disentangle shape and appearance:

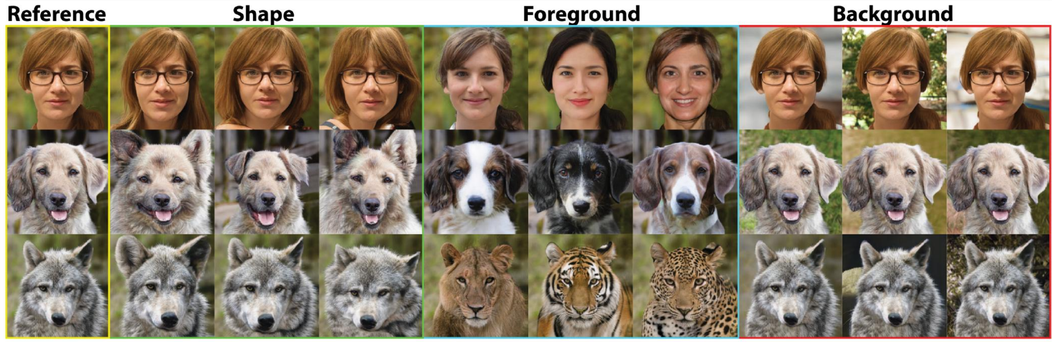
$$TOCS(\mathbf{r}) = \int_{t_n}^{t_f} w(\hat{\mathbf{r}}(t))\hat{\mathbf{r}}(t) \quad \text{with } w(\hat{\mathbf{r}}(t)) \text{ the SDF based weigth.}$$

Deferred Neural Renderer

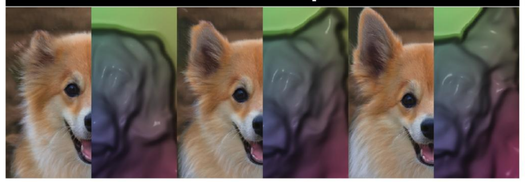


The 2D TOCS map conditions StyleGan2 blocks for the object appearance synthesis. Foreground and background are generated independently and composited for foreground/background **disentanglement**.

Disentangled control



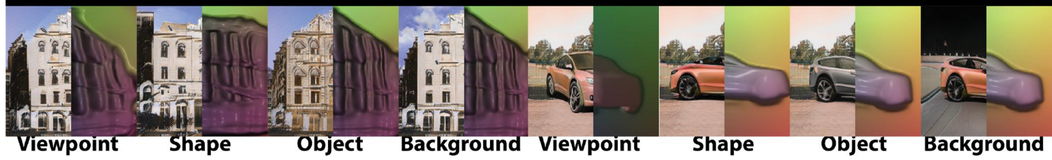
Deformation equivariance



Single-view reconstruction

Quantitative Baseline

		FFHQ 256² FID	AFHQ 256² FID			
		-	Cat	Wild	Dogs	Joint
Low-res	GRAF	71.00	-	-	-	121.00
	GIRAFFE	31.20	-	-	-	31.00
	pi-GAN	34.56	38.92	-	-	-
	Disentangled3D	28.18	-	-	-	-
Photorealistic	StyleNERF	8.00	-	-	-	14.00
	CIPS3D	6.97	-	-	-	-
	StyleSDF	11.50	-	-	-	12.80
	EG3D*	4.80	3.88	-	-	-
	GiraffeHD	11.93	12.36	-	-	-
	Ours	7.91	4.29	3.49	13.95	-

Qualitative Baseline

Limitations - Topology variation